

Snowledge: Use cases

Version history

Version	Date	Author	Description
0.1	28.10.2022	Sanni Onkalo	Add 3, 4, 5.1 and 5.2
0.2	1.11.2022	Sanni Onkalo	Add 1, 2, 3, 5.4, 5.5, 5.6
0.3	9.12.2022	Sanni Onkalo	Add 5.7 and minor changes to other use cases

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1. Purpose of this document

This is an appendix document of the Test plan for project Snowledge. The purpose of this document is to describe use cases that can be used in system testing and acceptance testing. This document also describes user stories and requirements that have been used as a helping tool in defining the use cases. Use cases are defined in part 5 of this document.

2. Dictionary

Name	Definition
System	Snowledge, consists of mobile app, web application and administrator side.
App	Snowledge Mobile app for iOS and Android devices

3. User stories

ID	Role	Why	What	Requirement ID
US01	App user	User is in mild emergency	User asks help from other users	RQ01
US02	App user	User is in serious emergency	User asks for help in serious emergency	RQ02
US03	App user	User sees another user in need of help	User helps other user in need of help	RQ03
US04	Administrator	App user is missing	User checks App user's past location points	RQ04
US05	App user	User is hiking in Pallas area during the winter	User checks current snow conditions in Pallas area	RQ05
US06	App user	User is hiking in Pallas area	User checks current weather conditions in Pallas area	RQ06

4. Requirements

ID	Requirement	Solution type	Solution ID	Solution
RQ01	Users can inform the app that they are in mild need of help	Feature	FE01	Asking for help in mild emergency
		Rule	RL01	User can ask for help from users that are in 3km radius. If no users are within 3km radius, this feature is not available.
RQ02	Users can contact National rescue service	Feature	FE02	Asking for help in serious emergency
		Rule	RL04	In serious emergency, users can ask for help from other users in 1km radius. If no users are in 1km radius, the help request is not sent to other users.
RQ03	Users can accept help requests from other users and see their location	Feature	FE03	Accepting a help request from other user
RQ04	Administrators can see App user's location history	Feature	FE04	Viewing App user's location history
		Rule	RL02	App user's current location is stored in System every 5 minutes when using the App.
		Rule	RL03	Location data is stored in the System for 4 days and then destroyed.
RQ05	App provides information on current snow conditions	Feature	FE05	Viewing current snow conditions in Pallas area
RQ06	App provides information on current weather	Feature	FE06	Viewing current weather conditions in Pallas area
RQ07	Users can easily turn off location sharing	Feature	FE07	Turning off location sharing

5. Use cases

5.1. Asking for help in mild emergency

Use case	Asking for help in mild emergency
Users	2 App users
Goal	Send a help request for other app users to see
Pre-condition	User 1 has allowed location sharing and User 2 is within 3km radius and has allowed location sharing and notifications
Post-condition	User 1 has sent a help request and User 2 can see the help request on their device
Normal activity	1. User 1 selects asking for help and App prompts user to select serious or mild emergency. 2. User 1 selects mild emergency and App prompts user for emergency type. 3. User 1 selects emergency type and App shows User 1's location on a map and informs the User that help request is active. 4. User 2 can see the help request on their device as a notification.
Alternative activity	3a. There are no other users near, in which case App informs User 1 of that.

5.2. Asking for help in serious emergency

Use case	Asking for help in serious emergency
Users	App user
Goal	Inform National Rescue Service and App administration of User's location in order to get help in a serious emergency
Pre-condition	User has allowed location sharing and User has 112 application on their device
Post-condition	User's help request has been sent to App administration and 112 application has been opened on User's device
Normal activity	1. User selects asking for help and App prompts user to select serious or mild emergency. 2. User selects asking for serious help. 3. 112 application opens on User's device and Help request is been sent to other app users in 1km radius.
Alternative activity	3a. User's device does not have 112 application, in which case User's device prompts User to call 112.

5.3. Accepting a help request from other user

Use case	Accepting a help request from other user
Users	2 App users
Goal	Help 2 App users find each other on a map in order for User 1 to help User 2
Pre-condition	User 1 has allowed location sharing and notifications
Post-condition	The help request has been accepted and User 1 and User 2 see each other's location on their own devices
Normal activity	1. User 2 does use case [FE01] Asking for help in mild emergency. 2. User 1 sees a notification on their device and opens the App 3. User 1 selects accepting the help request, in which case the App informs User 1 that their location is shared to User 2 and User 1 can see User 2's location on a map. 4. User 2 sees that their help request has been accepted and can see User 1's location on a map and distance to User 1.
Alternative activity	-

5.4. Viewing App user's location history

Use case	Checking App user's location history
Users	Administrator
Goal	Find out App user's past location points to help finding the App user
Pre-condition	System has at least one App user who has allowed location sharing and has been using the App for at least 4 days.
Post-condition	User has checked App user's location history successfully.
Normal activity	1.
Alternative activity	-

5.5. Viewing current snow conditions in Pallas area

Use case	Viewing current snow conditions in Pallas area
Users	App user
Goal	View current snow conditions in Pallas area
Pre-condition	System has data on current snow conditions in Pallas area.
Post-condition	User has viewed current snow conditions in Pallas area successfully.
Normal activity	1. User opens menu. 2. User selects Pallas snow conditions and App shows Pallas snow conditions.
Alternative activity	-

5.6. Viewing current weather conditions in Pallas area

Use case	Viewing current weather conditions in Pallas area
Users	App user
Goal	View current weather conditions in Pallas area
Pre-condition	System has data on current weather conditions in Pallas area.
Post-condition	User has viewed current weather conditions in Pallas area successfully.
Normal activity	1. User opens menu. 2. User selects weather information in Pallas and the App shows current weather information in Pallas area.
Alternative activity	-

5.7. Turning off location sharing

Use case	Turning off location sharing
Users	App user, Administrator
Goal	Turn off location sharing
Pre-condition	App user has allowed location sharing
Post-condition	App user has disallowed location sharing, and their location is no longer sent to the System.
Normal activity	1. User opens Location screen and App shows that location is allowed and the last time the location was sent. 2. User turns off location sharing, in which case no new location points are sent after that. 3. After 5 minutes, Administrator checks App user's location, and System shows a location point that was sent more than 5 minutes ago, and no newer location points.
Alternative activity	-